Lesson 5.2 Guided Notes

Session 1 · Area of Triangles

I can find the area of a triangle using one half the base times the height. · 6.G.A.1

Word / Spanish	Meaning — complete the blank	Example
right triangle / triángulo rectángulo	A triangle with one _____ corner.	its two legs are b and h
height / altura	The distance from the base to the opposite corner, at a _____ angle.	h in A = ½ × b × h
congruent / congruente	Exactly the same _____ and shape.	2 congruent triangles make a parallelogram

Key idea

Two congruent triangles make a _____.

So a triangle's area is _____ of base × height: A = ½ × b × h.

In a right triangle, the two _____ are the base and the height.

Worked example

A triangle has a base of 14 cm and a height of 9 cm. Find its area.

14 × 9 = _____

126 cm² is the area of the parallelogram made by two triangles.

126 ÷ 2 = _____

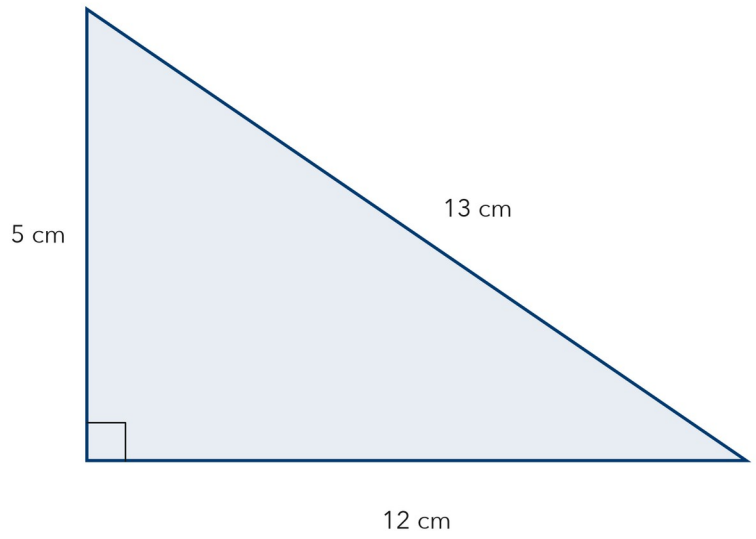
Answer: The area is _____ cm².

You try

1. Find the area of each triangle.
a. b = 10 m, h = 7 m _____ m²
b. b = 8 ft, h = 5 ft _____ ft²

_____

2. Find the area of the right triangle.



_____

Sum it up: The triangle formula has a ½ because ___.

_____

Session 2 · Missing Measures in Triangles

I can find a missing base or height of a triangle when I know its area. · 6.G.A.1

Word / Spanish	Meaning — complete the blank	Example
area / área	The space inside a shape, measured in _____ units.	40 cm²
formula / fórmula	A rule written with _____ and symbols.	A = ½ × b × h
inverse operation / operación inversa	An operation that _____ another.	doubling undoes taking half

Key idea

Undo the half first: _____ the area.

Then _____ by the measure you know.

A base or height is a length, so use units like cm, not _____.

Worked example

A triangle has an area of 40 cm² and a base of 10 cm. Find the height.

40 = ½ × 10 × h

Double the area: 2 × 40 = _____

Divide by the base: 80 ÷ 10 = _____

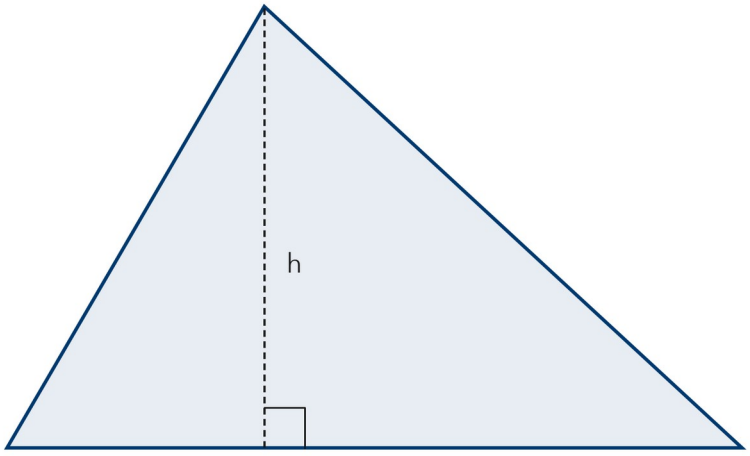
Answer: The height is _____ cm.

You try

1. Find the missing measure.
a. A = 30 ft², h = 6 ft; b = _____ ft
b. A = 21 m², b = 7 m; h = _____ m

_____

2. The area of the triangle is 48 m². Find the height h.



16 m

_____

Sum it up: To find a missing base or height, I double the area first because ___.

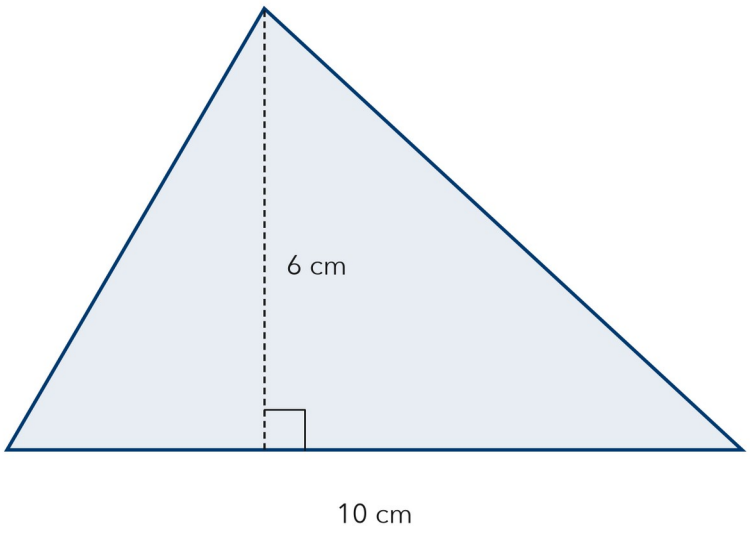
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Session 1 · Model, reason, and explain

Area of Triangles · 6.G.A.1

Success: represent the quantities, justify the relationship, and check the result with the correct units.

Two congruent triangles with base 10 cm and perpendicular height 6 cm form a parallelogram. Find one triangle area and explain the factor one-half.



_____

_____

_____

Extend your reasoning

Create a contrasting example that exposes the misconception: omits one-half or uses a nonperpendicular side.. Explain what changes and what stays true.

_____

_____

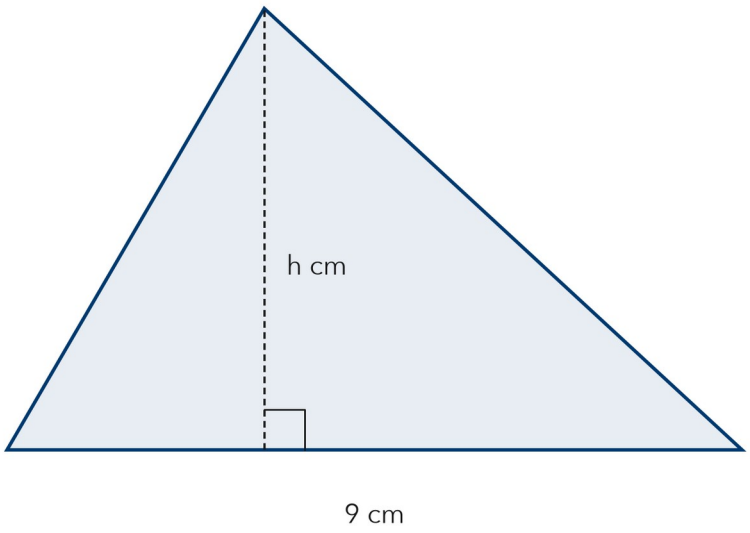
Self-check: Did my model, calculation, explanation, and units agree?

Session 2 · Model, reason, and explain

Missing Measures in Triangles · 6.G.A.1

Success: represent the quantities, justify the relationship, and check the result with the correct units.

A triangle has area 36 cm² and base 9 cm. Find the perpendicular height and check by substitution.



_____

_____

_____

Extend your reasoning

Create a contrasting example that exposes the misconception: divides area by base without accounting for one-half.. Explain what changes and what stays true.

_____

_____

Self-check: Did my model, calculation, explanation, and units agree?